

Efficient programming of a large-scale optical neural network with linear disordered media

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Abstract. Driven by artificial intelligence, the exponential growth in computational and energy demands has spurred exploration of alternative computing paradigms beyond traditional electronic processors. Optical neural networks (ONNs), a promising neuromorphic platform with light-speed computing, offer inherent advantages for accelerating artificial intelligence, including massive parallelism, ultralow latency, and reduced power consumption. However, most existing ONNs lack physical programmability after deployment due to rigid optical interconnects and limited optical transformations, which restrict their performance across diverse machine learning tasks. We propose a programmable and scalable ONN, an optoelectronic reservoir computing architecture integrated with cascaded linear disordered media. The key novelty lies in its physical kernel optimization via genetic algorithms, which involves programming diffuser orientations within a reduced search space and integrating wavefront shaping for feature preprocessing, thereby enhancing both optical random projection and preprocessing transformations. This platform achieves competitive accuracy in image classification with 93.5% fewer parameters than digital models; it also performs well in graph classification and human action recognition, proving feasibility for non-Euclidean and time-series datasets. Our work addresses the critical programmability bottleneck of ONNs, paving the way for scalable, energy-efficient ONNs with programmable physical kernels and opening new paths for high-performance physics-inspired neuromorphic computing in resource-constrained scenarios.

Keywords: optical neural network; deep learning; optical linear transformation; genetic algorithm; disordered media.

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1 Introduction

Artificial intelligence (AI) has profoundly transformed contemporary life, revolutionizing fields such as medical diagnosis, scientific imaging, robotics, and transportation.^{1–4} With this landscape, the emerging “AI for Science” paradigm is gaining traction across diverse scientific disciplines, driving an escalating demand for high-performance computing.^{5,6} Traditionally, silicon-based electronic computing hardware—built around transistor-based architectures—has been pivotal in advancing AI, enabling the training of increasingly complex and large-scale models.^{7–9} This progress has reinforced the long-held belief

that “bigger is better” in AI model development.¹⁰ However, the relentless scaling of electronic hardware has now encountered a developmental bottleneck, unable to keep pace with the exponential rise in computational demands.¹¹ This marks the onset of the post-Moore’s Law era, where traditional silicon-based scaling is no longer sufficient. Compounding this challenge is the inherent von Neumann bottleneck—the separation of memory and processing in conventional architectures—which severely limits performance scalability for large AI models.¹² As the computational cost of training state-of-the-art AI models continues to grow exponentially, the need for more efficient computing paradigms has become urgent.^{13,14} In response, sustainable efforts have been undertaken to develop machine learning (ML) accelerators that utilize a variety of technologies, including electronic and mechanical devices,^{15–17} photonic hardware,^{14,18} memristors,^{19,20} and even biological molecular reactions.^{21,22} Photonic

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neural systems, which leverage the rapid information-processing capabilities of light, enable neuromorphic computing with advantages such as massive parallelism, ultralow latency, and high energy efficiency.^{23,24} In recent years, optical neural networks (ONNs) have made significant advancements driven by the explosive growth of deep neural networks and an increasing demand for energy-efficient computing solutions.¹⁸ For instance, integrated photonic circuits^{25,26} have demonstrated on-chip parallel matrix multiplication capabilities in a highly compact and high-speed manner. Meanwhile, free-space light propagation through diffractive or convolution layers^{27–30} has been proven well for executing large-scale matrix operations.

Beyond conventional photonic systems that directly emulate digital neural networks, computation also arises naturally from complex light scattering within disordered optical media, an architecture known as optical reservoir computing (RC).^{31–35} The multiple scattering of light through random media inherently functions as a random projection kernel, which serves as a powerful computational primitive widely utilized in machine learning for dimensionality reduction and feature extraction in mathematics and statistics.³⁶ Unlike digital RCs that require the explicit storage of reservoir parameters, the disordered medium inherently serves as a reservoir of stochastic photonic connections, which are physically encoded in its scattering properties.³¹ As a result, the majority of computations occur within optical domains without explicit training, requiring only a simple digital readout layer—typically implemented via kernel ridge regression—for final decision. In addition to leveraging linear scattering properties, nonlinear optical effects have also been investigated to enhance the computational capacity of RC systems. Representative implementations include the nonlinear Kerr effects in multimode fibers^{37–40} and second harmonic generation occurring in lithium niobate nanostructures.⁴¹ However, these approaches typically necessitate high peak-power optical pulses to induce nonlinear responses. This may undermine the energy-efficient advantages of optical computing and could also result in suboptimal performance due to limited projection capacity. To address these limitations, several methods have been developed to achieve nonlinear processing using only linear optics. These strategies include utilizing intracavity multiscattering⁴² and implementing multilayer optical modulation,⁴³ thereby introducing structural nonlinearity. Notably, theoretical foundations such as the Johnson-Lindenstrauss Lemma⁴⁴ and Cover's Theorem⁴⁵ suggest that sufficiently high-dimensional linear random projection can exhibit significant expressive power in optical computing. When further integrated with programmable and scalable random projection kernels, such an optical reservoir computing system holds the potential to theoretically outperform the nonlinear processing strategies.^{42,43}

In this paper, we present a programmable and scalable optical neural network (PS-ONN) architecture that employs cascaded random projection layers as a promising computing paradigm. The proposed reservoir computing framework effectively shifts the majority of computational operations into the linear optical domain through controlled light propagation and scattering characteristics. To maintain system versatility across various ML applications while preserving intrinsic advantages of optical processing, we introduce a minimal digital decisive layer—eliminating the need for hardware reconfiguration. The PS-ONN architecture achieves classification accuracy comparable to fully digital models such as LeNet-5 while reducing trainable parameters by up to 93.5%. In comparison with other adaptive

ONNs, our programmable approach significantly diminishes optimization complexity by concentrating solely on tuning the angular combinations of cascaded disordered media via a genetic algorithm. This approach significantly diminishes optimization complexity, improves efficiency, and reduces training latency compared with approaches that require extensive weight optimization. Furthermore, to tailor the optical transformation within the physical medium, we employ a genetically optimized wavefront shaping (WFS) strategy^{46,47} at the optical front-end. This approach incorporates trainable parameters that refine optical transformation and further enhance the random projection capabilities, thereby improving overall network performance. This kernel programming approach has also been assessed in tasks involving non-Euclidean and sequential data structures—such as graph classification and human action recognition—demonstrating the versatility of the physical training method. Designed around the architecture of purely linear disordered media, this system provides both simplicity and compactness while maintaining ultra-low power consumption.^{48,49} This work presents a promising new approach for optical computing architectures, demonstrating valuable attributes across programmability, efficiency, and scalability that lay a foundation for advancing the performance of future optical computing systems.

2 Results

2.1 Experimental Setup and Principles

Figure 1 shows the schematic setup of our ONN, comprising data encoding, optical computing, and digital readout (see Sec. 4.1 in the [Appendix](#) for details). First, the data to be classified are encoded as phase patterns (see Sec. 4.2 in the [Appendix](#) for details) and subsequently loaded onto a 1030 nm continuous wave (CW) laser beam by a spatial light modulator (SLM). Then, the modulated light illustrates cascaded linear disordered media, thereby physically projecting input features into a high-dimensional latent space.⁴⁵ The motorized rotation stage enables tunable physical random projection kernels via controllable medium rotation. The implementation of cascaded kernels enhances projection capabilities and extends the optional parameter space compared with the single counterpart. Due to the unknown relationship between matrix distributions and angular configurations, the genetic algorithm (GA) is employed to optimize the projection kernel for performance, with each gene representing a set of rotation angles. The intensity of the optical speckle captured by the camera undergoes a quadratic nonlinear transformation.⁵⁰ These representations are then processed by a computationally efficient digital regression layer for various ML tasks (see the [Appendix](#) for data processing). This approach aligns with neuromorphic computing paradigms—such as reservoir computing⁵¹ and extreme learning machines^{52,53}—which integrate complex fixed connections with simple trainable post-processing. By utilizing accuracy as the objective function, we implement adaptive wavefront shaping to enhance the computational performance of the optical system. This is achieved through genetic programming of the preprocessing pattern, which optimizes either phase or amplitude modulation applied to the incident light. Such an approach facilitates precise control over the light scattering process induced by the diffuser, effectively tailoring the embedded optical computing mechanism. Consequently, this results in highly

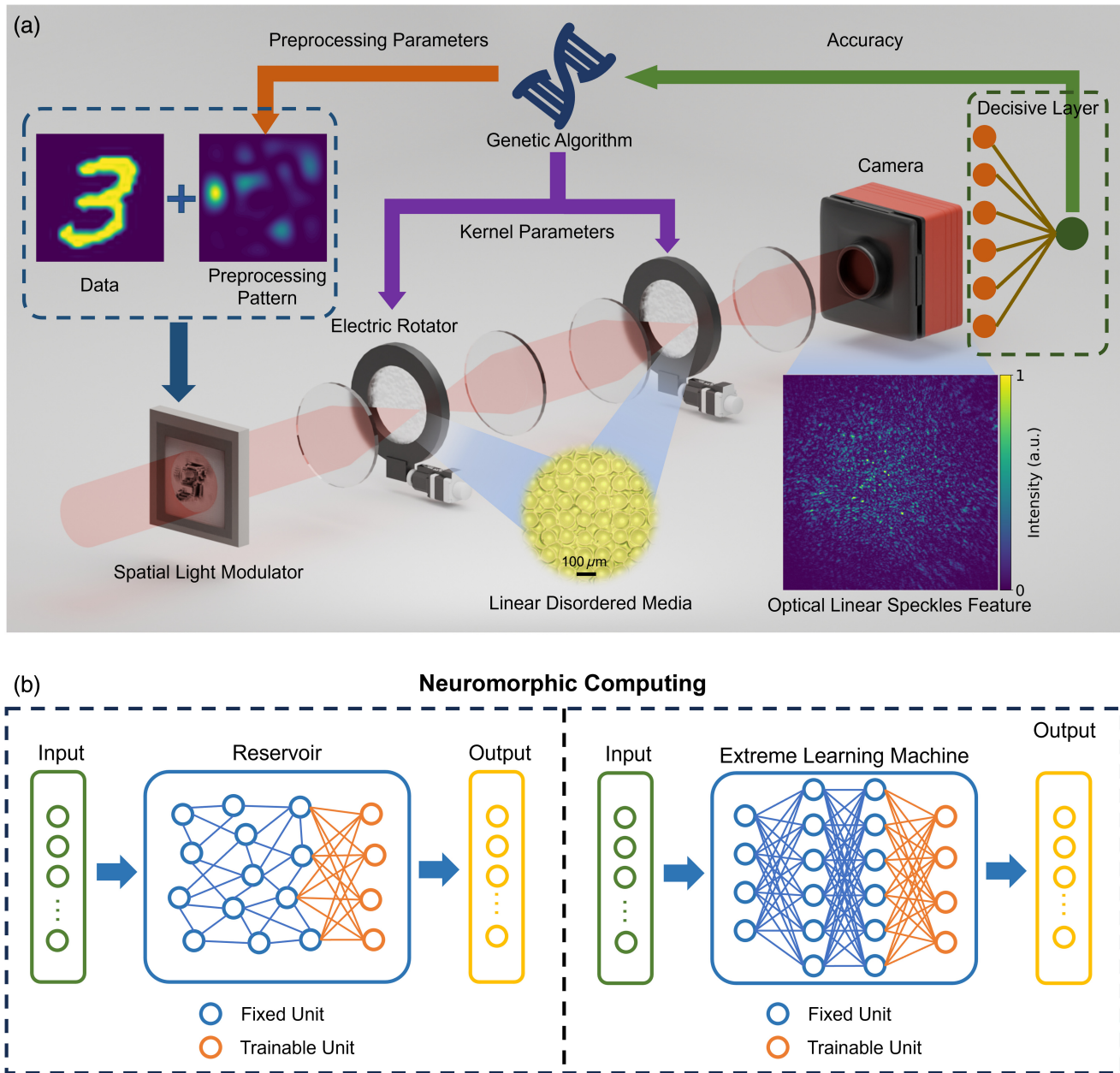


Fig. 1 Concept of the programmable and scalable optical neural network (PS-ONN). (a) The schematic setup for the PS-ONN. The input data of various formats are encoded onto the spatial phase of light with a preprocessing pattern superimposed. The modulated beam propagates through two linear disordered media and is then transformed into the optical features. These representations are then fed into a digital decision layer to make task decisions. Programming of kernels and WFS optical transforms is driven by the genetic algorithm. (b) Neuromorphic computing models (reservoir computing and extreme learning machine) implemented by the PS-ONN at the optical level. Blue and orange lines represent fixed and trainable connections, respectively.

discriminative projection patterns for subsequent interpretation. Experimental results demonstrate that significant performance improvements can be realized by tuning only a small number of key programming parameters. This finding underscores both the high sensitivity and computational expressiveness of the optical front-end in determining overall system performance. The performance of PS-ONN is assessed on various ML tasks, including image classification, graph classification, and human action recognition.

2.2 Image Classification Using Cascaded Physical Random Projection Kernels

To evaluate the utility of cascaded physical random projection kernels, we compare the performance of single- and dual-diffuser configurations for image classification tasks across three datasets with increasing complexity. As shown in Fig. 2, the sample subset is representative of the population and adequately explains the discrepancy. The first task is to classify MNIST

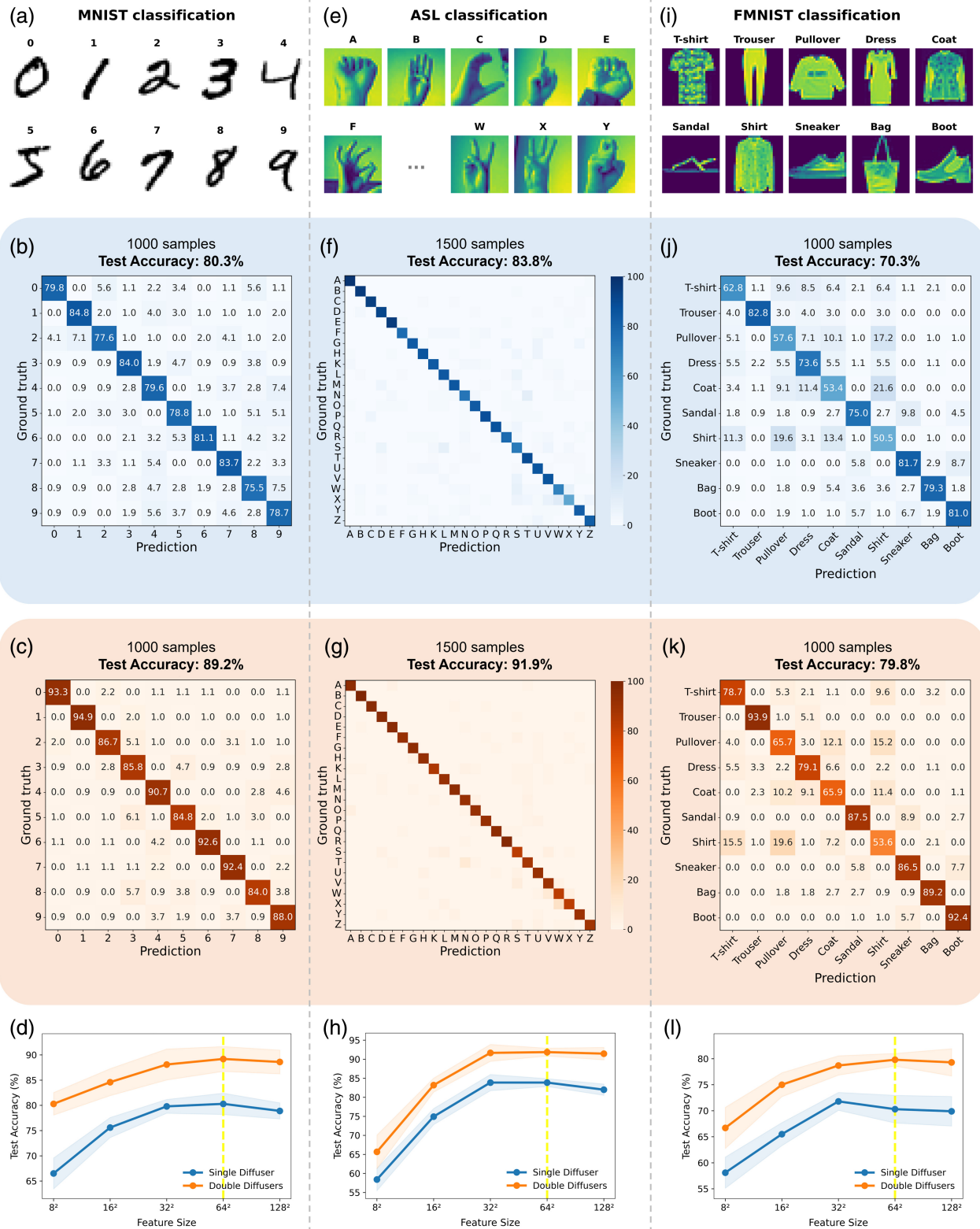


Fig. 2 Performance for image classification under single- and dual-kernel configurations. (a) 10-class MNIST-digit samples. (b), (c) Confusion matrices for single-diffuser (b) and dual-diffuser (c) configurations at a feature size of 4096. (d) Comparison of classification accuracies based on single diffuser (blue) and dual diffusers (orange) at varying feature sizes. The dual configuration outperforms its single counterpart with an accuracy increase ranging from 8.3% to 13.8%. (e) 24-class ASL alphabet samples. (f), (g) Same as panels (b) and (c) for the ASL alphabet dataset, (h), (i) Same as panels (b) and (c) for the FMNIST dataset, (j), (k) Same as panels (b) and (c) for the FMNIST dataset.

Fig. 2 (Continued) respectively. (h) Same as panel (d), with test accuracy improvement from 7.3% to 9.5% using dual kernels. (i) 10-class Fashion-MNIST samples. (j), (k) Same as panels (b) and (c) for the Fashion-MNIST dataset, respectively. (l) Same as panel (d), with test accuracy improvement from 6.9% to 9.5% using dual kernels. In panels (d), (h), and (l), shaded areas indicate ± 1 s.d. from fivefold cross-validation means.

handwritten digits⁵⁴ [10-class, Figs. 2(a)–2(d)]. To accommodate the SLM's effective encoding region (400 pixel $\times$ 400 pixel), each pixel from the original 28×28 image is magnified to a corresponding block on the SLM. Then, the optical features recorded by the camera are used for digital classification. Figures 2(b) and 2(c) display the confusion matrices obtained from single- and dual-kernel configurations, with test accuracies of 80.3% and 89.2% by respective ridge regression layers with 4096 optical feature nodes. Our analysis validates the concrete and universal advantage of the dual-kernel architecture, demonstrating that cascaded configurations outperform single-diffuser implementations by 8.3% to 13.8% across all feature dimensions, with optimal accuracy achieved at a moderate feature size of 4096. All of them are presented as the mean $\pm$ one standard deviation (s.d.), derived from five-fold cross-validation. Then, we proceed to the 24-class American Sign Language (ASL) alphabet dataset⁵⁵ [Figs. 2(e)–2(h)], using a balanced subset of 1500 samples for evaluation. The cascaded kernel speckles demonstrate more expressive features with a test accuracy of 91.9%, compared with 83.8% obtained with a single-kernel implementation. In addition, we further evaluate our photonic computing system on the more challenging Fashion-MNIST dataset⁵⁶ [Fig. 2(i)], which is more representative of real-world vision tasks. As shown in Figs. 2(j) and 2(k), the cascaded system delivers an absolute accuracy improvement of 9.5% at a dimension of 4096. Experimental results on the Fashion-MNIST dataset show consistent accuracy gains across various feature sizes, underscoring the importance of the enhanced random projection capability enabled by cascaded kernels in boosting performance.

2.3 Physical Model and Numerical Study of PS-ONN

For clarity, the random projection performed by PS-ONN can be investigated by the beam propagation method (BPM), which can be mathematically expressed as

$$I = |P_3 \cdot T_2 \cdot P_2 \cdot T_1 \cdot P_1 \cdot (E_0 \cdot e^{iD})|^2, \quad (1)$$

where P describes the propagation distance and T is the spatial transmission matrix of the diffuser. E_0 denotes the initial optical amplitude. The input data D is then transformed to optical representations I captured by a camera (see Sec. 4.3 in the [Appendix](#) and Note 3 in the [Supplementary Material](#) for details). The free-space propagation can be modeled using the angular spectrum method.⁵⁷ Notably, beyond the nonlinearities introduced by phase encoding and intensity detection, the random projection achieved by pure linear disordered media attains performance comparable to that of the nonlinear random projection case.⁴¹ This can be attributed to the cascaded kernel's enhanced projection capability, as further validated by the following numerical results on the MNIST-digit dataset. An example of an image loaded onto the SLM is shown in Fig. 3(b). The transmission matrices of dual diffusers are modeled as randomized phase screens [Figs. 3(c) and 3(d)], whereas the resulting normalized speckle features captured by the camera are depicted in

Fig. 3(e). These optical representations are then taken as the final feature vector for classification. Figure 3(f) shows an example of the prediction and ground truth, demonstrating great agreement. The confusion matrices demonstrate a significant 9% accuracy improvement for the dual-diffuser configuration [Fig. 3(h)] over the single-diffuser counterpart [Fig. 3(g)]. Figure 3(i) compares experimental and simulation results, showing an accuracy degradation of $\sim 1\%$ in the experiment relative to numerical simulations. This discrepancy is likely attributable to practical noise sources such as laser intensity fluctuations and camera readout noise. Additional scaling performance of PS-ONN is provided in Note 5 and Fig. S4 in the [Supplementary Material](#).

2.4 Programming Kernels for Image Classification

The random projection performance of PS-ONN is determined by the selection of physical kernels (see Sec. 4.4 in the [Appendix](#)), whereas the elusive relationship between kernel matrix distributions and angular configurations creates an opportunity for optimization-based kernel discovery. Setting classification accuracy as the fitness metric, cascaded kernels are genetically programmed on a representative subset of 1000 samples, reducing the required computation by nearly two orders of magnitude. Solutions are encoded by 10-bit binary strings, representing angular positions discretized into 1024 steps over 360 deg, with each string defining a rotation angle pair for the dual stages. The GA evolves eight generations from an initial population of six candidates and evaluates 54 unique angle pairs. The zero-angle configuration serves as a baseline for assessing optimization effectiveness.

The kernel programming schedule begins with the MNIST-digit dataset [Fig. 4(a)]. As shown in Fig. 4(b), the progression of fitness yields a 3.6% accuracy improvement, as evidenced by the comparison of confusion matrices before [Fig. 4(c)] and after [Fig. 4(d)] optimization. Then, we proceed to the ASL dataset [Fig. 4(e)]. To accommodate the 24-class classification, we enlarge the subset scale and expand the GA population to facilitate broader exploration. The accuracy gain of 2.4% is achieved, as shown in Fig. 4(g). When scaling the sample number to 5000, the optimized system attains 99.12% accuracy. Further assessment is conducted on the Fashion-MNIST dataset [Fig. 4(i)], presenting a more challenging classification task than MNIST-digit. Despite the increased difficulty, our programming method still delivers 3.0% accuracy improvement, raising performance from 79.8% to 82.8% [Fig. 4(j)]. As shown in Figs. 4(k) and 4(l), confusion matrices show consistent accuracy gains across all categories, with the notable improvements in commonly confused classes such as shirts, pullovers, and dresses. Moreover, we challenge our photonic neural system with more demanding tasks. Figures 4(m)–4(p) demonstrate the experimental results for the CIFAR-10 database.⁵⁸ An accuracy improvement of 3.9% (37.0% versus 33.1%) is attained at a feature size of 4096 [Figs. 4(o) and 4(p)]. Using the optimized kernels, we evaluate the classification performance on the full CIFAR-10 dataset. The resulting

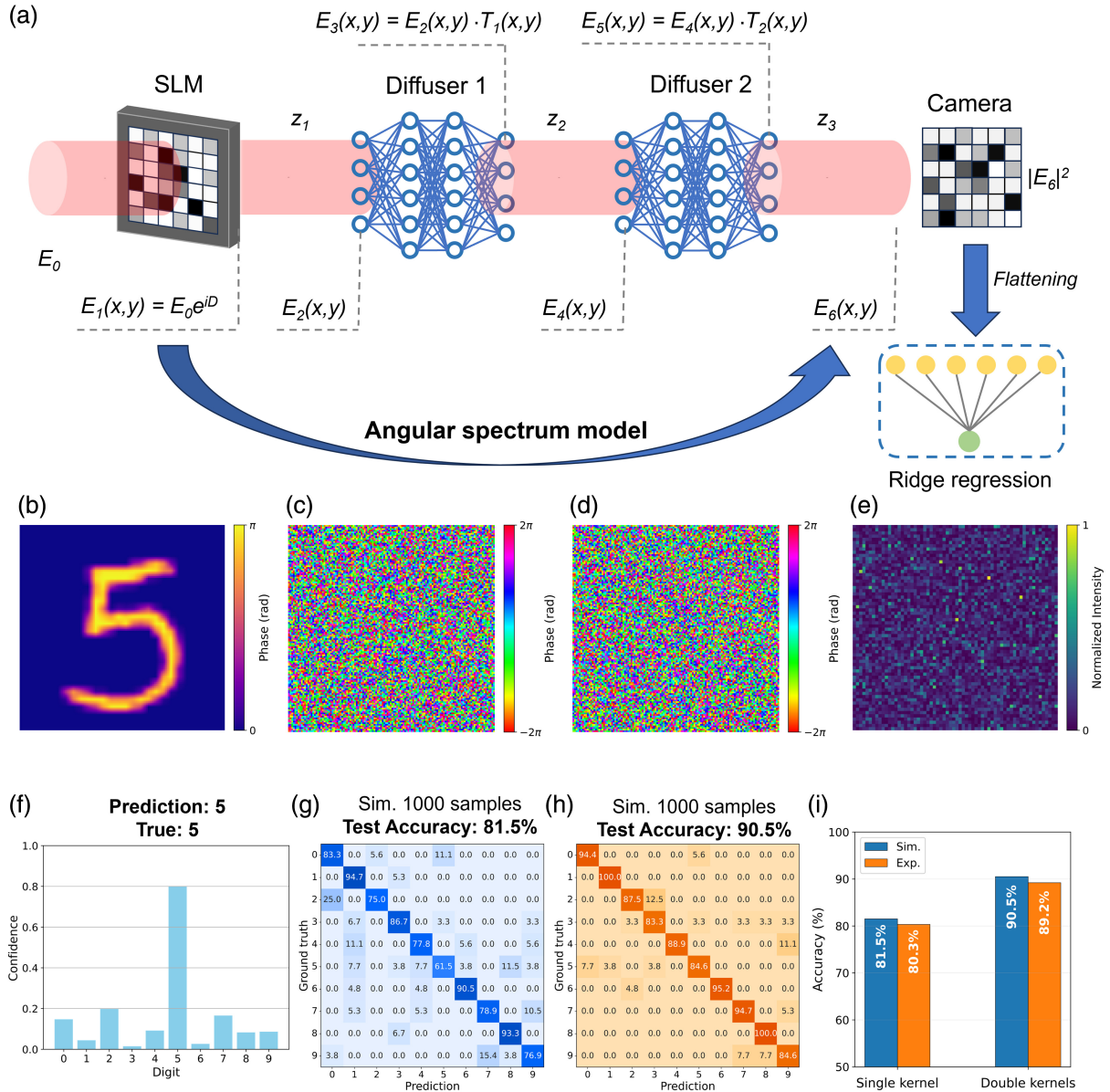


Fig. 3 Physical model and numerical results of the PS-ONN. (a) Schematic of the PS-ONN system. Samples are phase-encoded by an SLM, with light propagation described via the angular spectrum method. The optical speckle features are flattened and then classified using a ridge regression classifier. (b) Phase pattern of an MNIST digit. (c), (d) Phase transfer matrices of two diffusers obtained through numerical simulations. (e) Simulated speckle representations. (f) Classification confidence scores from ridge regression for all digit classes. (g), (h) Confusion matrices for single-diffuser (g) and dual-diffuser (h) configurations, revealing improved accuracy with cascaded kernels. (i) Experimental versus simulated accuracy comparison for both configurations, demonstrating close agreement ($<1.5\%$ difference).

accuracy of $\sim 55\%$ surpasses that of classic digital convolutional networks such as LeNet-5.⁵⁴ The observed improved accuracy in classifying diverse datasets demonstrates the robust effectiveness of this programmable kernel method.

2.5 Wavefront Shaping Approach for Programmable Optical Transformation

To optimize the optical transformation for effective interpretation of optical features at the decisive layer, we employ a WFS

approach to integrate a fixed and pre-programmed pattern into the data encoding process (see Sec. 4.4 in the Appendix). The WFS pattern is genetically optimized to enhance performance in the specific ML task, thereby introducing trainable optical parameters at the preprocessing stage. After programming kernels, the random projection capability can be further enhanced by employing programming WFS for programmable optical transformation. The combination of wavefront shaping and data encoding can be formalized as follows, where the complex encoded value on the SLM $E_{\text{SLM}}^k(x,y)$ is given by

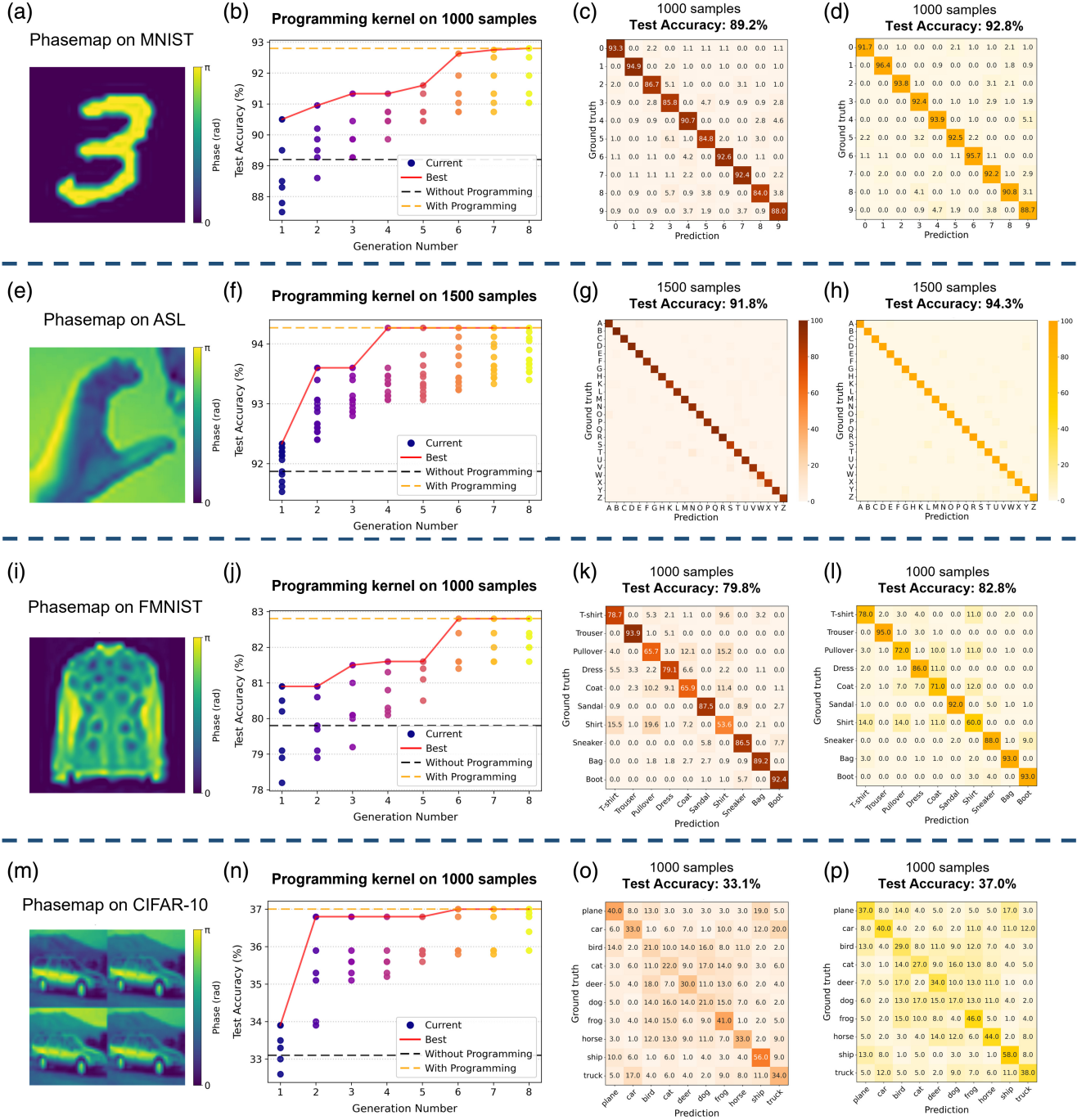


Fig. 4 Efficient programming of kernels for image classification. (a) MNIST-digit sample loaded onto SLM. The phase map is obtained based on grayscale values. (b) Genetic programming progress with 3.6% accuracy improvement. (c), (d) Confusion matrices before (c) and after (d) optimization. (e) ASL alphabet sample. (f) Same as panel (b), showing 2.4% accuracy improvement. (g), (h) Same as panels (c) and (d) for the ASL alphabet dataset. (i) FMNIST sample. (j) Same as panel (b), demonstrating 3.0% accuracy improvement. (k), (l) Same as panels (c) and (d) for the FMNIST dataset. (m) CIFAR-10 RGB sample. (n) Same as panel (b), exhibiting 3.9% accuracy improvement. (o), (p) Same as panels (c) and (d) for the CIFAR-10 RGB dataset.

$$\begin{aligned}
 E_{\text{SLM}}^k(x, y) &= WF(x, y) \exp(iD^k(x, y)) \\
 &= |WF(x, y)| \exp(i(D^k(x, y) + \text{Arg}(WF(x, y)))).
 \end{aligned}
 \quad (2)$$

A phase-only SLM is enough to apply both the amplitude and phase modulation in the WFS pattern. The former $|WF(x, y)|$ can be realized by modifying the strength of a blazed grating pattern, whereas the latter $\text{Arg}(WF(x, y))$ can be

directly encoded as a phase pattern along with input data $D^k(x, y)$. In this work, a phase-only WFS pattern is programmed to demonstrate proof-of-concept optical transformation optimization. To further enhance the input data's

response to WFS-mediated optical transformation, we introduce an optional lightweight computational data feature augmentation module at the digital end that is accessible, which can be readily deployed on edge detection devices in the future.

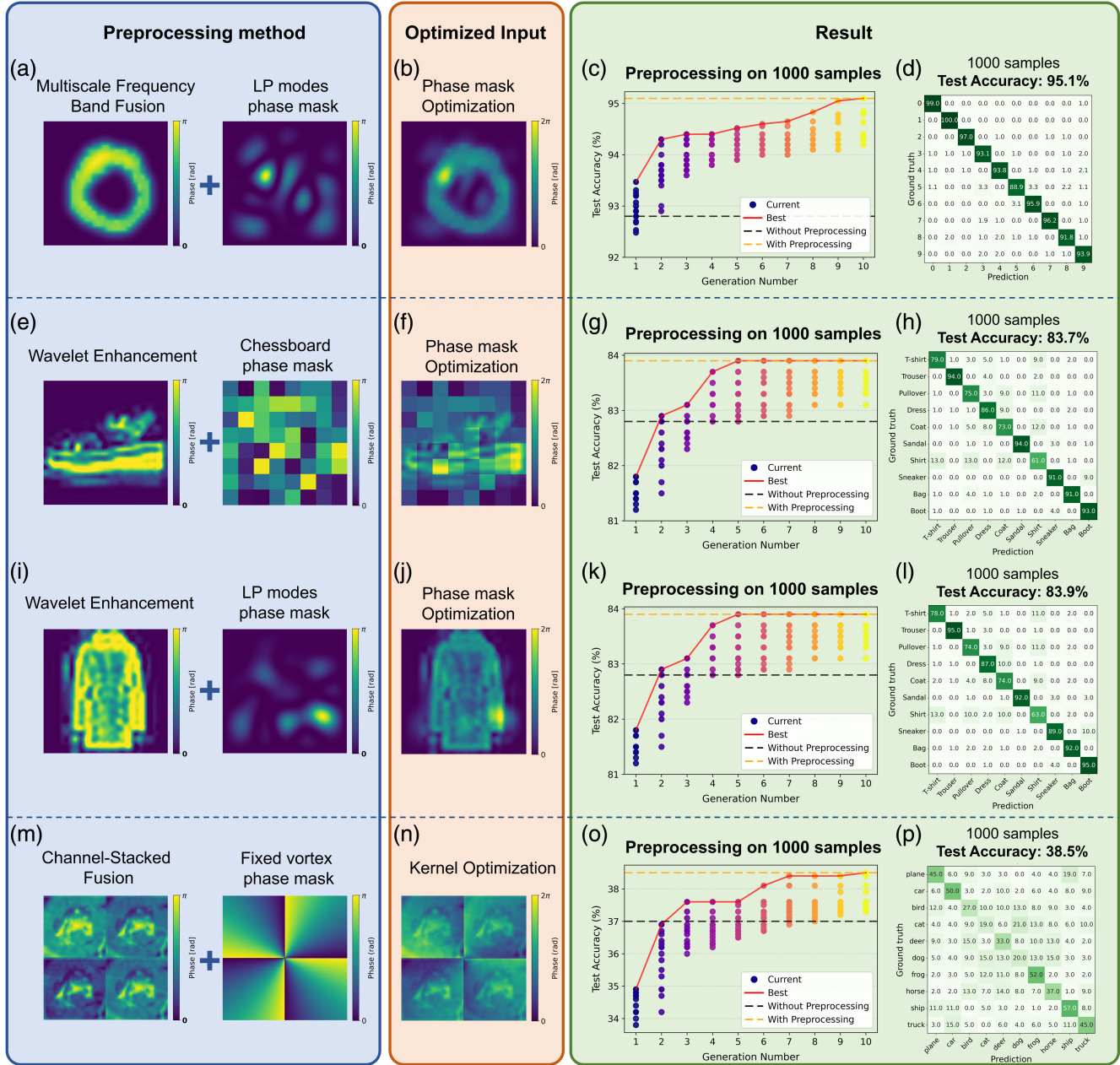


Fig. 5 Efficient programming of WFS optical transform for image classification. (a) WFS preprocessing procedure on the MNIST-digit dataset. The multiscale frequency band fusion sample is superimposed onto the LP-mode phase map. (b) Optimized programmed input pattern on the SLM. (c) Performance improvement of genetic programming for WFS optical transform, achieving a 2.3% accuracy gain. (d) Confusion matrix with optimized preprocessing. (e), (i) WFS preprocessing procedure on the Fashion-MNIST dataset. Wavelet-enhanced samples are superimposed with a chessboard (f) or LP-mode (j) phase map. (g), (k) Same as panel (c), with accuracy improvements of 0.9% and 1.1%, respectively. (h), (l) Confusion matrices for both optimized configurations. (m) WFS preprocessing procedure on the CIFAR-10 dataset. The channel-stacked fusion sample is superimposed with a truncated vortex phase map. (n) Kernels are reprogrammed with a superimposed structured field. (o) Same as panel (c), yielding a 1.5% improvement. (p) Optimized confusion matrix.

The programmable WFS approach is first evaluated on the MNIST-digit classification task (see the [Appendix](#)). For feature enhancement, we adopt the multiscale frequency band fusion strategy to reconstruct a more robust and informative representation. The WFS pattern is generated from an optimized combination of the first 30 linearly polarized (LP) modes in a multimode fiber. This reduces the WFS pattern optimization to determining merely 30 weighting coefficients. The augmented representation and the initial WFS phase pattern are demonstrated in Fig. 5(a). The WFS data's pattern is genetically optimized using a population size of 15 over 10 generations on a subset of 1000 samples. The optimized input and the fitness progression are shown in Figs. 5(b) and 5(c), respectively. This optimization yields an accuracy improvement of 2.3%, achieving a final accuracy of 95.1% [Fig. 5(d)]. The system achieves a classification accuracy of 97.7% on the full dataset, closely approaching the 98.3% performance of a digital LeNet-5 model while utilizing 93.5% fewer parameters. Table 1 compares the classification accuracies of various optical or digital approaches on the MNIST-digit dataset. Then, we test on the Fashion-MNIST dataset (see the [Appendix](#) and Table S2 in the [Supplementary Material](#)), and the results are shown in Figs. 5(e)–5(l). The wavelet-based enhancement technique⁵⁹ is adopted, with the WFS pattern utilizing the chessboard or LP mode configurations [Figs. 5(e) and 5(i)]. Experimental results show consistent accuracy gains, and the confusion matrices [Figs. 5(h) and 5(l)] validate robust classification across most categories. Despite employing only a linear random projection, our system achieves an overall accuracy of 89.4%, surpassing the 88.5% accuracy of the digital LeNet-5 implementation. Moreover, we challenge the WFS approach on the CIFAR-10 dataset (see the [Appendix](#) and Table S3 in the [Supplementary Material](#)), with the channel-stacked fusion approach employed to process the RGB images. Unlike previous processing procedures, our system leverages the constant helical phase profile as the WFS pattern [Fig. 5(m)], employing the vortex beam as a spatial filter to extract frequency components.⁶⁰ Subsequently, the kernels are reprogrammed to achieve an optimal projection with the fixed WFS pattern imposed [Fig. 5(n)]. The fitness progression demonstrates a 1.5% improvement in accuracy [Fig. 5(o)], supported by the corresponding confusion matrix in Fig. 5(p). The optimized system achieves a test accuracy of 55.26% on the full CIFAR-10 dataset, demonstrating the viability of optical processing for complex visual recognition tasks. Comparative performance analysis with state-of-the-art ONNs is provided in Table S4 in the [Supplementary Material](#).

2.6 Programming Kernels for Graph Classification

Graph-structured data represents a fundamentally different paradigm from conventional Euclidean data. Although traditional

datasets (e.g., images) typically exhibit vector-like or grid-like structures, many real-world entities are fundamentally defined by their interconnections. These relationships naturally lend themselves to graph representations, where entities correspond to nodes and their interconnections form edges. Such graph-structured data have become ubiquitous across diverse domains, ranging from molecular chemistry (where molecules are modeled as atomic graphs) to social network analysis (where individuals and their relationships form complex networks). To effectively leverage information from these non-Euclidean structures, many approaches have been proposed to learn representations from non-Euclidean graph data, including frequent subgraph mining,⁶¹ graph convolutional network,⁶² and graph kernels.⁶³ At the core of graph analysis lies the dual representation of both structural and attribute information: the graph topology is formally captured through adjacency matrices, whereas node-specific features are represented via feature matrices. This dual formalism enables comprehensive analysis of both relational patterns and node attributes in graph-structured data.

To process graph data with a similar number of nodes, our photonic neural system directly encodes the graph structure in the form of an adjacency matrix onto an SLM, facilitating subsequent optical representation extraction. Figures 6(b)–6(e) present the experimental results on a two-category graph dataset generated using the stochastic block model (SBM).⁶⁴ All graphs contain a constant number of nodes (60 in this study), with the classification difficulty dictated by the interclass similarity k (closer to 1 means a greater similarity between two classes). For visualization, we reduce feature dimensions from 4096 to 2 using the uniform manifold approximation and projection (UMAP), revealing the cluster distributions before and after programming kernels [Figs. 6(b) and 6(c)]. As shown in Fig. 6(d), the classification accuracy improves by 4.5% for $k = 1.3$, a trend that remains consistent across varying interclass similarities [summarized in Fig. 6(e)]. Moreover, we evaluate on a real-world Reddit-binary (RB) graph database⁶⁵ [Figs. 6(f)–6(i)]. A key challenge in encoding these graphs stems from the wide variation in their sizes, from as small as six nodes and four edges to as many as 3728 nodes and 4071 edges. Inspired by the graphlet kernel-based approach that quantifies structural similarities through subgraph comparison,⁶⁶ we develop a generalized graph processing workflow, as illustrated in Fig. 6(a). Given a large graph, we first sample multiple subgraphs with a constant number of nodes and extract their corresponding adjacency matrices. For subgraphs with insufficiently sampled nodes, we employ empty node padding. Subsequently, the resulting representations are aggregated by averaging to produce a final feature vector of each graph, which is then passed to the digital classifier (see the [Appendix](#)). The task is to distinguish

Table 1 Comparison of accuracies among different approaches in the classification of the MNIST-digit dataset.

Approach	Single diffuser	Dual diffusers	2 diffusers with kernel optimization	2 diffusers with preprocessing optimization		Ridge classifier	LeNet-5
Input	Optical	Optical	Optical	Optical	Optical	Digital	Digital
Samples	1000	1000	1000	1000	70,000	70,000	70,000
Accuracy	80.3%	89.2%	92.8%	95.1%	97.7%	86.1%	98.3%
Parameters	4096	4096	4096	4096	4096	4096	61706

Note: Entries in bold indicate the performance of the proposed ONN method on the whole MNIST-digit dataset for clarity.

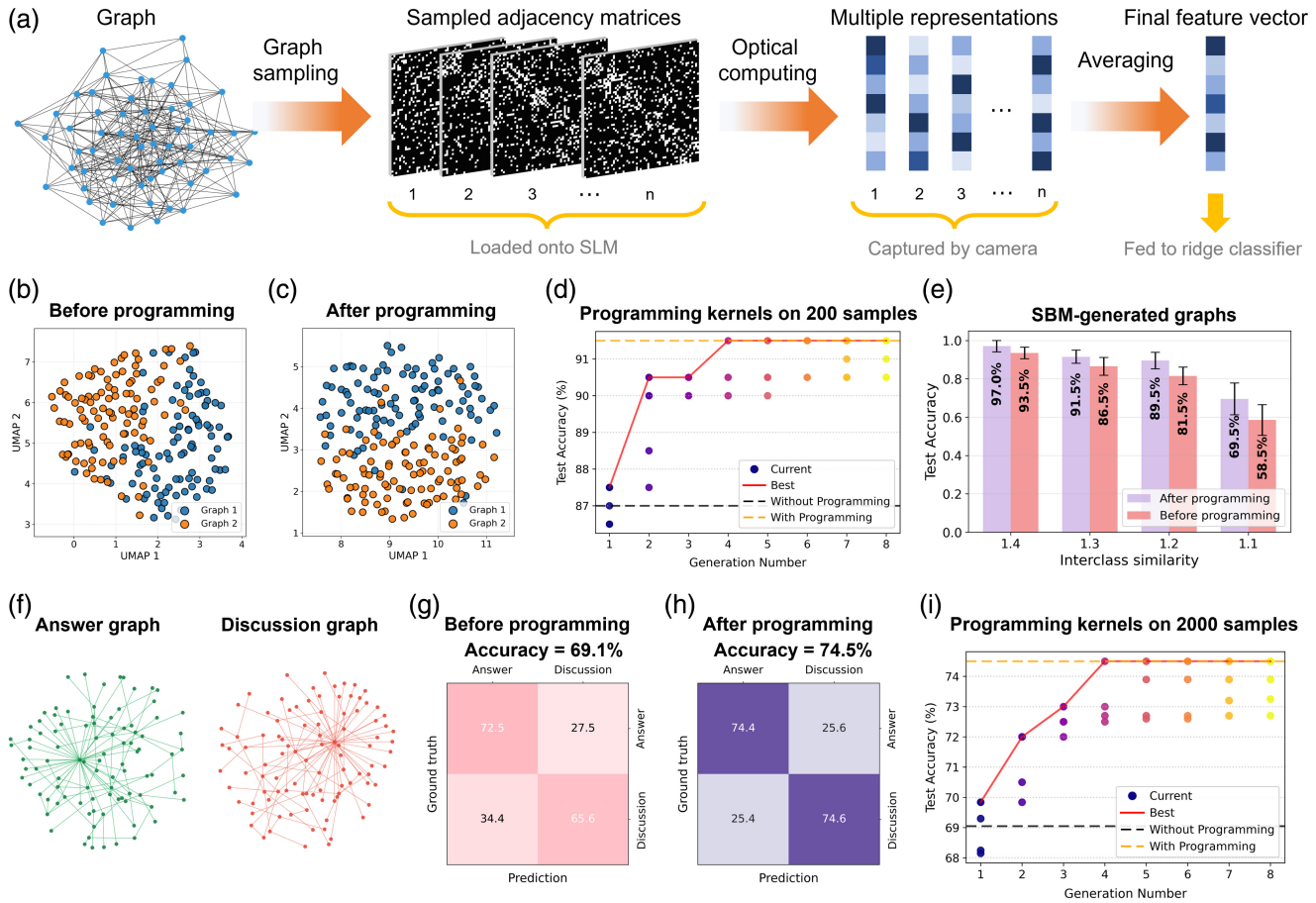


Fig. 6 Efficient programming of kernels for graph classification. (a) Processing flow of graph classification on the PS-ONN. Adjacent matrices are repeatedly sampled from a graph, and their corresponding speckle representations are optically extracted. The average of multiple representations is taken as the final feature vector for classification. (b), (c) Visualizations of UMAP results for SBM-generated graph data (interclass similarity $k = 1.3$) obtained before programming (b) and after programming (c) at a feature size of 4096. (d) Accuracy progression of genetic programming for graph classification, with an improvement of 4.5% for $k = 1.3$. (e) Bar chart of cross-validation test accuracies before and after programming at varying interclass similarities. (f) Illustration of discussion graph and answer graph samples from the RB dataset. (g), (h) Confusion matrices on the test set obtained before programming (g) and after programming (h) at a feature size of 4096. (i) Same as panel (d), with test accuracy improvement of 5.4% after genetic programming.

between “discussion graph” and “answer graph” [Fig. 6(f)]. As illustrated in Figs. 6(g)–6(i), programming the kernels leads to improved classification accuracy, confirming the capability of PS-ONN to process graph-structured data.

2.7 Programming Kernels for Human Action Recognition

In addition to processing static images, the PS-ONN is also capable of handling temporal sequence data, and this capability is critical for analyzing dynamic systems. To illustrate its functionality, we conduct a comprehensive time-series classification task using human action recognition from video sequences, which is a representative time-varying data modality. Specifically, we conduct the classification task using the Weizmann database,⁶⁷ which comprises video recordings of 10 distinct motions (bend, jack, jump, pjump, run, side, skip,

walk, wave1, and wave2) performed by nine subjects. As shown in Fig. 7(a), we adopt a sliding window segmentation technique to decompose the entire and redundant video streams into shorter video clips. Each clip is assigned labels corresponding to the original video sequence so that models can focus on learning mappings from these shorter and overlapping clips. Each generated clip inherits the label of its source sequence, enabling the models to learn meaningful mappings from temporal segments to action categories. Video frames within each temporal clip are sequentially loaded onto the SLM. Subsequently, the optical features extracted from all frames within a clip are concatenated to create a comprehensive representation. These clip-level representations, which maintain the temporal coherence of the original sequences, are subsequently fed into a digital ridge classifier for final action recognition. The overall sequence accuracy is determined by statistically summarizing the inference results from all clips in the test set through the application of a

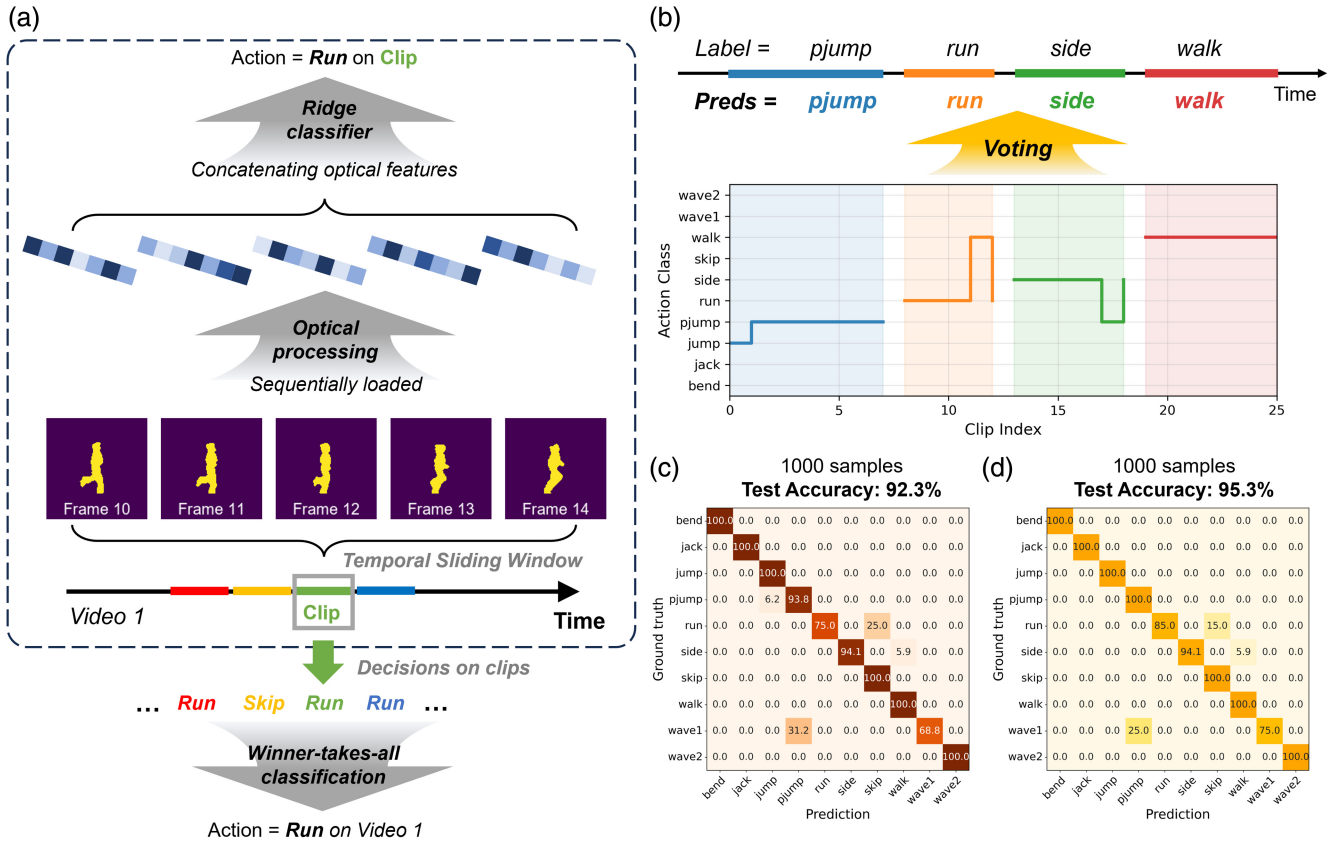


Fig. 7 Efficient programming of kernels for human action recognition. (a) Procedure for video-based human action recognition. A sliding window technique segments the video sequence into several shorter clips, each containing an equal number of frames that are sequentially loaded onto the SLM to extract corresponding optical features. The features from each clip are concatenated and subsequently input into the digital ridge classifier. For interference, the final decision is determined using a winner-takes-all strategy. (b) Demonstration of various prediction results on video clips within the same sequence. (c), (d) Confusion matrices on the test set obtained before programming (c) and after programming (d) at a feature size of 4096.

winner-takes-all strategy [Fig. 7(b)]. Before the programming of kernels, the PS-ONN had already achieved a relatively high accuracy of 92.3%, thereby demonstrating the effectiveness of this time-series data processing schedule. As illustrated in Figs. 7(c) and 7(d), the implementation of programming kernels results in an accuracy improvement of 3.0%.

3 Discussion and Conclusion

To demonstrate the optical advantage in computing speed and energy efficiency, we present a comprehensive analysis of the PS-ONN's computational performance [see Fig. 3(a) and Note 2 in the [Supplementary Material](#) for details]. Currently, the system operates at 50 Hz due to SLM speed limits, delivering 21.8 TFLOPS and 0.41 TeraOP/J. By integrating digital micromirror devices (DMDs)—which offer significantly faster modulation speeds—and a custom-designed electronic classifier, we anticipate achieving a throughput of 436 TFLOPS and 253.6 TeraOP/J. These projected specifications represent a 62× increase in computational throughput and a 939× improvement in energy efficiency compared with NVIDIA's V100 GPU (Table S1 in the [Supplementary Material](#)). Furthermore, PS-ONN matches LeNet-5's classification accuracy while reducing parameters

by 93.4% on MNIST-digit/Fashion-MNIST and 96.2% on CIFAR-10. These results establish a promising paradigm for energy-efficient, high-throughput computing using optical physical computation. Detailed comparisons with ridge classifier and LeNet-5 digital models, as well as linear discriminant analysis (LDA) results, are provided in Notes 2–4 and Figs. S2 and S3 in the [Supplementary Material](#). The detailed comparative analysis of optical speckle patterns and random projection kernels, together with reproducibility evidence, is provided in Notes 6 and 7 and Figs. S5–S10 in the [Supplementary Material](#).

Compared with state-of-the-art optical neural networks—such as D2NN,^{27–30} linear optical reservoir computing,³¹ nonlinear reservoir computing^{37–40} and nPOLO⁴²—our proposed architecture demonstrates several significant advantages. For example, in classification tasks involving MNIST and Fashion-MNIST datasets, our programmable cascaded-kernels architecture utilizes only linear disordered media as random projection kernels, yet it outperforms other competing ONNs that rely on nonlinear projections or more complex architectures. This superior performance is achieved by programming the kernels and WFS optical transformation to amplify the capabilities of linear random projection—an approach aligned with the objective of the nPOLO method, which also seeks to leverage linear optics to achieve

or even exceed nonlinear-level performance. Similar to other advanced optoelectronic systems, our PS-ONN effectively integrates the parallelism and power efficiency inherent in photonics with the flexibility characteristic of lightweight electronic systems. This integration facilitates a wide range of ML tasks—including complex domains such as graph classification and human action recognition—without necessitating hardware modifications. Furthermore, unlike systems reliant on precise power control (e.g., ONNs using multimode fibers or nonlinear materials), PS-ONN maintains robust performance at ultralow continuous-wave power levels. This capability positions it favorably for potential applications in few-photon optical computing and quantum information processing.^{48,49}

In summary, the PS-ONN represents a high-performance optoelectronic architecture that integrates optical random projection with digital intelligence. By leveraging cascaded linear disordered media as scalable analog feature extractors, the system significantly reduces digital computational overhead and reveals the cutting-edge concept of Physics-Inspired Computing. Through the genetic programming of random projection kernels and WFS optical transformations, our framework achieves adaptive and task-specific feature extraction with remarkable efficiency. This physically programmable architecture demonstrates versatility across various ML applications and can also be extended to nonlinear random projection systems—thereby paving the way for more powerful and reconfigurable optical neural networks. This physical programmability extends naturally to emerging domains such as generative photonic computing⁶⁸ by optimizing the optical intrinsic noise sampler, an area of growing research interest. Results and analyses confirm that PS-ONN matches the accuracy of digital neural networks while delivering substantial improvements in processing speed and energy efficiency improvements exceeding several orders of magnitude compared to GPU-based systems. By utilizing passive optical components for large-scale matrix operations, this system achieves exceptional computational throughput at ultralow power consumption. We posit that the PS-ONN bridges neuromorphic computing principles and photonic advantages, thereby establishing a robust framework that merges optical parallelism with electronic programmability. This advancement paves new pathways for highly scalable, energy-efficient intelligent systems as well as future high-performance optical computing endeavors.

4 Appendix: Methods

4.1 Experimental Setup

Our experimental system utilizes commercially available optoelectronic devices as the building modules of PS-ONN. The energy source for optical computing is generated by a continuous-wave, linearly polarized single-frequency laser operating at a wavelength of 1030 nm, with output power set to 5 mW. The coherent light wavefront is collimated and expanded using lenses before being directed to a reflective phase-only SLM. This SLM comprises 1024 pixel $\times$ 1024 pixel capable of modulating the incident wavefront at a maximum speed of 60 Hz with 8-bit accuracy. The modulated light is relayed optically with a 4*f* system that includes two relay lenses, after which it is focused by another lens before reaching the scattering medium. The scattering medium is constructed from N-BK7 glass substrate embedded with precisely arranged polymer microstructures designed to suppress the zero-order diffraction while

ensuring uniform scattering characteristics. Each microstructure unit features a circular geometry with well-defined dimensions (90 μm diameter $\times$ 30 μm thickness) and exhibits $\sim 90\%$ optical transmittance at a wavelength of 1030 nm. The medium is mounted on a motorized rotation stage to facilitate angular control. As light propagates through cascaded layers with a scattering medium, speckle patterns are captured optically by a camera. Before the optical features are captured, a 1000 $\times$ absorption-type attenuator is employed, reducing the power incident on the CMOS sensor to 5 μW . The camera exposure time is set to 8 ms and is synchronized with the refresh cycle of the SLM. Throughout these experiments, camera parameters—including exposure time, saturation levels, and gain—are maintained constant. The resulting optical features are subsequently processed for various machine learning tasks using ridge regression on an external host computer; programming kernels and WFS preprocessing were executed automatically within the closed-loop system. The overall experimental system stability can be found in Note 1 and Fig. S1 in the [Supplementary Material](#).

4.2 Data Processing

We utilize a phase-only SLM to encode input data into optical fields. In terms of data normalization, two distinct approaches are applied at different stages of the experiment: (1) Min-Max normalization is applied to the images, which are linearly scaled from the original 0 to 255 intensity range to a phase range of 0 to π . (2) Z-score normalization is applied to the captured optical features before they are processed by the ridge regression model. The central region of the SLM (around 400 pixel $\times$ 400 pixel) is designated for data encoding, whereas other pixels remain unchanged. For grayscale image datasets such as MNIST-digit, Fashion MNIST, and ASL, we directly display each input image on the SLM. By contrast, for color image datasets such as CIFAR-10, we concatenate three channels of each image (i.e., red, green, and blue) along with its grayscale version to form four blocks in a single composite image. The fivefold cross-validation technique is employed in image classification to ensure robust evaluation. For SBM-generated and Reddit-binary graph datasets, we encode the adjacency matrix of the graph onto the SLM. This study focuses exclusively on undirected graphs and binary adjacency matrices that only contain connection information; however, it should be noted that our proposed system is not limited to these cases. All graphs generated from the SBM consist of 60 nodes; consequently, their corresponding adjacency matrices are sized at 60 pixel $\times$ 60 pixel and are displayed directly on the SLM. For the RB database featuring varying numbers of nodes and edges, we use the random walk approach to sample each original graph with a sample size of 30 nodes over 80 times. Subsequently, we group these 80 sampled adjacency matrices into 20 distinct phase masks—each comprising four matrices. In instances where subgraphs do not meet the required number of nodes, empty nodes are introduced. These phase masks are sequentially loaded onto the SLM to generate 20 speckle features, which are then averaged together to create a final graph representation utilized for the following digital readout process. Given the limited sample size, we use a 16:9 data splitting strategy and perform five repeated experiments to obtain the final results. The Weizmann human action database comprises 93 video sequences featuring nine

subjects, each of whom performs ten distinct actions. Each video sequence contains ~ 30 to 120 frames captured at a frame rate of 50 frames per second (fps), with a resolution of 144×180 . Silhouette binary masks of the subjects are centered and cropped to dimensions of $130 \text{ pixel} \times 130 \text{ pixel}$ and subsequently up-sampled to 400×400 to align with the input size. The video sequences are organized in a ratio of 16:9 based on the number of subjects involved. To expand the effective training sample size, we use a sliding window technique that decomposes each video sequence into several shorter clips. In our experiment, we utilize an overlapping ratio set at 0.7 and define the length of each clip as 15 frames. Consequently, this approach yields a total of 1202 video clips along with their corresponding sequence labels. Frames within the same clip are sequentially loaded onto the SLM. The resulting multiple optical features are concatenated to form representations for each respective clip. During the training stage, the model is trained using optical features extracted from individual video clips. A winner-takes-all approach is used for classifying each complete video sequence. The classifier output is evaluated across all clips within one sequence; thus, the final classification result corresponds to the class associated with the majority representation among those clips attributed to that particular sequence.

These optically extracted features are subsequently input into a digital readout layer designed for machine learning tasks. It implements a straightforward linear regression model, which is trained using the Tikhonov regularization method. To facilitate this, we organize the N -dimensional feature vectors from K samples into a matrix denoted as X . The primary goal is to identify a linear combination of the elements within the feature vector that can most accurately predict each corresponding label, which is grouped in a vector y . Specifically, we aim to adjust the weight matrix W to minimize the l_2 norm of the residuals, expressed as $\|XW - y\|_2^2$. To penalize large weight values, an additional regularization term $\alpha \|W\|_2^2$ is incorporated into the objective function. This approach enhances the model's generalization performance and mitigates the risk of overfitting. Consequently, we derive an analytical solution for the readout matrix as $\hat{W} = (X^T X + \alpha I)^{-1} X^T y$. During experiments, we constantly set α at a moderate value of 100 across all cases to ensure fair comparisons. This practice contributes to maintaining strong data consistency and computational stability.

4.3 Physical Model and Simulations

The physical model of our PS-ONN can be characterized through four fundamental physical processes. (1) Input encoding. Input samples are converted to a phase map and then encoded onto an SLM, loading data onto the phase of light. (2) Diffuser modeling. Under the thin-element approximation, each diffuser is represented by a spatial-domain transmission matrix that accounts solely for phase modulation (intensity modulation is negligible). This facilitates projection of the input data into a high-dimensional latent space via element-wise matrix multiplication. (3) Free-space propagation. Light propagation through free space and optical components is accurately simulated using the angular spectrum method (ASM). (4) Nonlinear transformation. The nonlinearity of the system arises from two primary sources: the phase encoding nonlinearity associated with the SLM and the quadratic nonlinearity inherent in optical intensity detection by the camera. This configuration is commonly found in most architectures of ONNs.

Regarding the optical computing process, the input data $D(x, y)$ is first phase-encoded, producing a modulated optical field $E_1(x, y) = E_0 \cdot e^{iD(x, y)}$. The modulated light propagates from the SLM to the first diffuser, modeled by the ASM as $E_2(x, y) = \mathfrak{F}^{-1}\{\mathfrak{F}\{E_1(x, y)\} \cdot e^{i\phi(f_x, f_y, z_1)}\}$. It then interacts with the first scattering medium, resulting in $E_3(x, y) = E_2(x, y) \cdot T_1(x, y)$. Subsequently, the field propagates to the second diffuser, expressed as $E_4(x, y) = \mathfrak{F}^{-1}\{\mathfrak{F}\{E_3(x, y)\} \cdot e^{i\phi(f_x, f_y, z_2)}\}$ and undergoes the second scattering, which gives $E_5(x, y) = E_4(x, y) \cdot T_2(x, y)$. Finally, the light propagates to the detector plane, leading to $E_6(x, y) = \mathfrak{F}^{-1}\{\mathfrak{F}\{E_5(x, y)\} \cdot e^{i\phi(f_x, f_y, z_3)}\}$ and detected by the camera, $I_{\text{dec}}(x, y) = |E_6(x, y)|^2$. Summarizing above, the complete optical computing system implements a nonlinear transformation expressed as $I_{\text{out}} = O(I_{\text{in}})$, where I_{in} and I_{out} represent the input data and output feature vector, respectively, and O denotes the composite optical transformation including nonlinear phase encoding, free-space propagation, scattering effects, intensity detection, etc. (see Note 3 in the [Supplementary Material](#) for more details).

In the simulation, the wavelength is set to 1030 nm. The spacing between the two diffusers is maintained at 20 cm, whereas both the distance from the SLM to the first diffuser and from the second diffuser to the camera are kept constant at 10 cm. To enhance computational efficiency, the resolutions of the SLM and camera are configured as 128×128 and 64×64 , respectively. The model's performance is evaluated on 1000 samples from the MNIST-digit dataset to report test accuracy. The scattering behavior of each diffuser is modeled using a random phase screen generated from the Gaussian independent distribution, which is parameterized by correlation length and strength; these parameters dictate both the spatial frequency of phase variations and the intensity of the scattering modulation, respectively.

4.4 Programming Procedures of Random Projection Kernels and Wavefront Shaping Optical Transformation

As for programming of random projection kernels, we use the genetic algorithm to search for optimal kernels tailored to specific tasks. To generate distinct random projection kernels, we position the scattering medium at offset positions from the rotation axis. The GA explores the angular configuration space, which is discretized into 1024 equally spaced positions spanning 360 deg. The fitness metric can be defined by either accuracy or mean square errors on test samples, depending on the tasks at hand. Typically, we use performance at the zero-degree position as a reference point. For each candidate angle pair, performance is evaluated based on the optoelectronic hardware response before precisely adjusting the diffusers' angular position via a computer-controlled rotation stage for subsequent evaluations. For standard 10-class classification tasks (MNIST, FMNIST, CIFAR-10), we employ a population size of 6 across eight generations. For 24-class ASL, we scale the parameters to 15 and 10, respectively. For SBM-generated and RB graph classification and Weizmann human action recognition, we evaluate using a population size of 4 over eight generations. Both input data and WFS patterns are phase-encoded within a range from 0 to π . This approach enables phase superposition patterns remain within the modulation range of the SLM.

To further enhance the random projection capability, we program a wavefront shaping optical transformation for image classification as a demonstration. Initially, we apply data feature enhancement techniques tailored to various datasets. For the MNIST-digit dataset, we implement multiscale frequency band fusion. Low-, mid-, and high-frequency features are extracted using Gaussian blur, difference-of-Gaussian, and Canny edge detection methods, respectively. Given that low-frequency features play a dominant role in MNIST-digit classification, we empirically integrate them with weighting coefficients of 0.7, 0.2, and 0.1; these coefficients can be refined further. For classification on the Fashion-MNIST dataset, we develop a wavelet-based enhancement technique that transfers high-frequency features into the low-frequency domain. This approach is motivated by the observation that scattering media used for random projections perform better in the low-frequency range than in the high-frequency range. The input images undergo decomposition via db2 wavelet transforms; subsequently, high-frequency information is amplified along horizontal, vertical, and diagonal directions with weights of 0.4, 0.2, and 0.6, respectively. This amplified information is then combined with low-frequency data before undergoing an inverse wavelet transform to yield an enhanced input characterized by integrated high- and low-frequency features. For the CIFAR-10 dataset, we implement a channel-stacked fusion approach that combines both chromatic and luminance information by integrating original RGB channels with an additional grayscale channel to create an enhanced composite input structure measuring 2 pixel $\times$ 2 pixel per channel. Following feature enhancement procedures across all datasets mentioned above, we program the WFS pattern to facilitate optical transformation aimed at improving preprocessing outcomes. Both LP-mode-based and chessboard-based phase masks are proposed to generate flexible phase distributions while reducing the optimization parameter space. The LP-mode phase pattern is derived from the combination of intensity distributions of the first 30 orthogonal fiber modes obtained from a multimode fiber. For the chessboard phase pattern, each 50 pixel $\times$ 50 pixel block serves as a macro-block that is optimized within a much smaller parameter space. We demonstrate an alternative WFS processing strategy on CIFAR-10. Rather than optimizing the WFS phase pattern, we employ a fixed helical phase mask to introduce orbital angular momentum to the incident wavefront. This approach aids in separating information of different spatial frequencies, thereby facilitating the extraction of optical features. The employed phase mask utilizes a topological charge of 2 within $[0, \pi]$. Subsequently, we reprogram the random projection kernel under structured illumination conditions. For our genetic algorithm, we set the population size and number of generations to 15 and 10, respectively, allowing for increased search attempts.

Disclosures

The authors declare no competing interests.

Code and Data Availability

Data underlying the results presented in this paper may be obtained from the authors upon reasonable request.

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